BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 037 · Operating Systems: Interfaces and Memory

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. After marking an assessment, name one error category you should identify.

Q2. Which is malware?
A. Ransomware
B. A trusted security update
C. A spreadsheet formula
D. An authorised backup

Q3. Given A has code 65 and capital-letter codes are consecutive, give the code for C and explain how you found it.

Q4. A website needs English, Greek and Chinese text. “Unicode has fewer characters than standard ASCII.” Correct this. Why does Unicode suit this site?

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Operating system — System software managing hardware/resources and supplying services to programs.

GUI — Graphical User Interface using windows, icons, menus and a pointer.

CLI — Command Line Interface using text commands.

Memory management — Allocation and management of RAM for running programs.

Multitasking — Coordinating the execution of multiple applications.

User interface — Means by which the user interacts with a computer.

RAM — Volatile primary memory holding active programs and data.

New learning

Why an operating system?: System software coordinates hardware and applications, and provides services such as a user interface, memory management and security. Without it, applications would have to manage many hardware details.

Graphical and command interfaces: A GUI lets users select icons, windows and menus, making common tasks approachable. A CLI accepts typed commands, which can be efficient for an experienced user and automation.

Allocating memory: The OS allocates available RAM to applications, manages which data is kept in RAM and moves data as needed. Paging and segmentation are not required for this specification.

Multitasking: The OS manages running applications and shares processor time; it allocates memory and handles resources so users can switch between tasks. Switching does not mean one CPU core necessarily runs every task at exactly the same instant.

Worked example

A student has a browser, music app and editor open. Explain how the OS manages their memory and allows multitasking. [4 marks]

1. Identify the three applications competing for resources.

2. The OS allocates space in RAM to applications [1], keeping their required data available [1].

3. The OS manages CPU time among tasks [1], so the student can switch between or use different applications [1].

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

Q2. OS allocates RAM to each program [1] so needed data is available [1]. It manages the running tasks and processor time [1], so the user can switch between programs while both remain active [1].

Q3. GUI uses visual windows/icons and menus [1], making routine functions easier to discover [1]. CLI takes typed commands [1] that can be efficient for expert repetitive tasks [1]. OS allocates RAM to programs [1], allowing applications to run with required data in memory [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Missing knowledge, a wrong technical term, a missing cause and effect, or a missing detail.

Q2. A — Ransomware

Q3. 67; C is 2 position(s) after A.

Q4. Unicode supports far more characters than standard ASCII. It can cover English, Greek and Chinese. Use an encoding with enough codes; Unicode characters do not always take one byte.

I Do / We Do pages · Guided practice

Q1. The OS allocates available RAM to each application [1], allowing program data to be held in primary memory [1]. It manages CPU time or resources [1], letting both applications remain usable through multitasking [1].

Q2. A CLI can run repeated or scripted commands efficiently [1], speeding up frequent technical tasks [1]. Users must remember accurate commands [1], so beginners may find it harder and make errors [1].

Q3. The OS must allocate the available RAM across programs [1] to prevent applications from competing for the same allocated space [1]. It manages moving/loading application data in memory [1], so programs have needed data as the user switches between them [1].

You Do pages · Code First

```
freeMB = 1200
appMB = 850
print(freeMB - appMB)
```

Expected result: 350

You Do pages · Independent questions

Q1. Provides a user interface [1]; allocates/manages RAM for applications [1]. Other valid functions include file, device, user management and multitasking.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

4. Four linked marks: memory allocation, purpose, processor time, multitasking.

We Do · Practise with guidance

Guided 1

A receptionist runs booking software and an email client together. Explain TWO ways the OS helps these applications operate. [4 marks]

Worked method: The OS allocates available RAM to each application [1], allowing program data to be held in primary memory [1]. It manages CPU time or resources [1], letting both applications remain usable through multitasking [1].

Guided 2

A technician prefers typed commands for repetitive setup tasks. Explain one benefit and one drawback of a CLI compared with a GUI. [4 marks]

Worked method: A CLI can run repeated or scripted commands efficiently [1], speeding up frequent technical tasks [1]. Users must remember accurate commands [1], so beginners may find it harder and make errors [1].

Guided 3

A computer has three programs open but limited RAM. Explain TWO memory-management problems the OS must handle. [4 marks]

Worked method: The OS must allocate the available RAM across programs [1] to prevent applications from competing for the same allocated space [1]. It manages moving/loading application data in memory [1], so programs have needed data as the user switches between them [1].

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict free RAM after loading the application (MB).

```
freeMB = 1200
appMB = 850
print(freeMB - appMB)
```

predict

Question 1

State TWO functions of an operating system. [2 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

Question 2

A photographer switches between photo-editing software and a browser. Explain how operating-system memory management and multitasking support this. [4 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

Question 3

Compare GUI and CLI interfaces for an office worker and explain the OS role in memory management. [6 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

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BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

Exit Ticket · Answer before checking

The operating system can allocate RAM to different applications to help them run.